Rewarding Learning

ADVANCED
General Certificate of Education
2025

Digital Technology

Assessment Unit A2 1

assessing

Information Systems

[ADT11]

FRIDAY 13 JUNE, AFTERNOON

MARK SCHEME

General Marking Instructions

Introduction

Mark schemes are published to assist teachers and students in their preparation for examinations. Through the mark schemes teachers and students will be able to see what examiners are looking for in response to questions and exactly where the marks have been awarded. The publishing of the mark schemes may help to show that examiners are not concerned about finding out what a student does not know but rather with rewarding students for what they do know.

The Purpose of Mark Schemes

Examination papers are set and revised by teams of examiners and revisers appointed by the Council. The teams of examiners and revisers include experienced teachers who are familiar with the level and standards expected of students in schools and colleges.

The job of the examiners is to set the questions and the mark schemes; and the job of the revisers is to review the questions and mark schemes commenting on a large range of issues about which they must be satisfied before the question papers and mark schemes are finalised.

The questions and the mark schemes are developed in association with each other so that the issues of differentiation and positive achievement can be addressed right from the start. Mark schemes, therefore, are regarded as part of an integral process which begins with the setting of questions and ends with the marking of the examination.

The main purpose of the mark scheme is to provide a uniform basis for the marking process so that all the markers are following exactly the same instructions and making the same judgements in so far as this is possible. Before marking begins a standardising meeting is held where all the markers are briefed using the mark scheme and samples of the students' work in the form of scripts. Consideration is also given at this stage to any comments on the operational papers received from teachers and their organisations. During this meeting, and up to and including the end of the marking, there is provision for amendments to be made to the mark scheme. What is published represents this final form of the mark scheme.

It is important to recognise that in some cases there may well be other correct responses which are equally acceptable to those published: the mark scheme can only cover those responses which emerged in the examination. There may also be instances where certain judgements may have to be left to the experience of the examiner, for example, where there is no absolute correct response – all teachers will be familiar with making such judgements.

AVAILABLE MARKS

1 (a) Network card

It connects a computer to a LAN
… so that it can communicate with the file server
It provides a port
… for connecting a network cable/Ethernet cable
3 × [1]

Server

Manages network resources
... such as storage
... files
... hardware devices/peripherals/printers
… communications/email
3 × [1]

Repeater

Regenerates data transmission signals
… that have been attenuated
… by travelling over long distances
This increases the network's range
3 × [1] [9]

(b) There is no dedicated/central server
All computers have equal status
Each computer is a supplier of resources
… and a consumer of resources
4 × [1] [4]

(c) Only authorised users can log on
… using a (valid) username and password
Usernames and passwords can be managed centrally
… or users may be required to change passwords regularly
Log maintained of logons/attempted logons
… to prevent unauthorised access/hacking
4 × [1] [4]

AVAILABLE MARKS

(d) Banded response

Bus Network

A main cable/backbone to which all nodes are connected. Terminators are positioned at both ends. A node sends a data packet to another node by placing it on the backbone.

Star Network

A central or dedicated/host/fileserver/hub. Each node is directly connected to the hub by its own cable. The fileserver controls all network communication. The fileserver controls the transfer of data packets.

Evaluation wrt data security

The star network is more secure. Data packets are sent to/from the fileserver from/to the node directly via a dedicated connection. The fileserver controls all security. Only the intended recipient receives the data packet. The bus network is less secure. There is no central control of security. All nodes have access to the backbone/all nodes could access the data packet.

Level	Marking Criteria	Marks
Band 2	The candidate • Provides a detailed and accurate description of both topologies • Provides an explicit evaluation both of topologies, with justification, with respect to data security • Uses the appropriate Digital Technology terminology accurately throughout the response Presentation, spelling, punctuation and grammar are of a high standard.	[5]–[6]
Band 1	The candidate • Provides a detailed description of both topologies or discusses both topologies with respect to data security • Uses some relevant Digital Technology terminology Presentation, spelling, punctuation and grammar are sufficiently competent to make the response clear.	[3]–[4]
Band 0	The candidate • Provides a description of one topology which lacks detail or refers to one topology with respect to data security • Makes limited use of Digital Technology terminology Presentation, spelling, punctuation and grammar are such that the intended meaning is not completely clear.	[1]–[2]

[6] 23

AVAILABLE MARKS

2 (a) (i) Purpose of Application set

Acts as an interface between user applications and the network
Provides resources/services/data for user applications

Purpose of Transport set

Manages network traffic/ensures accurate delivery of data packets
2 × [1] [2]

(ii) Session

(A session begins) when an application wants to connect to a remote server
This layer opens a temporary communication channel between them
It synchronises the exchange of data/controls access
Interacts with the presentation and transport layers
3 × [1]

Network

Manages the routing of data packets
Reads the address of incoming packets
… and forwards them to the correct recipient
… using the optimum route
Interacts with the transport and data link layers
3 × [1] [6]

(iii) Any HW/SW component which meets part of the OSI standard
… will work with any other HW/SW component meeting the same part of the standard
Manufactures develop HW/SW components for a specific layer
… and these automatically work with HW/SW components designed for the layer above
… and the layer below
The OSI model is not dependent on a particular operating system
… and is implemented on a range of operating systems
4 × [1] [4]

(b) CSMA/CD

Monitors the traffic on a data channel/detects when a device wishes to transmit data
A collision occurs when two network devices attempt to use a data channel simultaneously
… after a random time interval
… the two devices attempt to transmit again
If another collision occurs, the time interval is increased
4 × [1]

RFID

An RFID tag contains a small computer chip
… programmed with (identifying) data
The tag is activated by radio waves from an RFID reader
The tag sends the data stored in its memory back to the reader
The tag can be active or passive
4 × [1] [8]

AVAILABLE MARKS

(c) Banded response

Parity bits

A specific bit in a byte is the parity bit/the /MSB in a byte is the parity bit
The number of 1's in a byte is odd if odd parity is being used/even if even parity is being used
When a byte is received its parity is checked
If the parity of the byte has changed, an error has occurred

Evaluation: How a parity bit will react to each error

Error One

The parity of the byte has changed from odd to even so an error will be detected

Error Two

The parity of the byte is still odd so the error will not be detected as an even number of bits have flipped

Parity checking only

Errors involving an odd number of bits flipping will be detected but which bits have flipped will not be known

Level	Marking Criteria	Marks
Band 2	The candidate • Provides a detailed description of a parity bit which is correct • Explains clearly with justification how a parity bit deals with both errors • Uses the appropriate Digital Technology terminology accurately throughout the response Presentation, spelling, punctuation and grammar are of a high standard.	[5]–[6]
Band 1	The candidate • Provides a detailed description of a parity bit which is correct • Accurately describes how a parity bit will react to one of the errors • Uses some relevant Digital Technology terminology Presentation, spelling, punctuation and grammar are sufficiently competent to make the response clear.	[3]–[4]
Band 0	The candidate • Provides a description of a parity bit which is correct but lacks detail **or** Refers to how a parity bit can detect an error • Makes limited use of Digital Technology terminology Presentation, spelling, punctuation and grammar are such that the intended meaning is not completely clear.	[1]–[2]

[6] 26

AVAILABLE MARKS

3 **(a)** Data is stored in a number of tables
Each table has a primary key
… which is a unique reference for each record/ which identifies each record in the table
The tables are linked together by relationships
… on common/shared fields
Relationships can be 1:1/1:m/m:m
Data can be retrieved from a number of tables using a single query
Data redundancy is minimised
4 × [1] [4]

(b) **Entity**
An object
… about which data is stored
2 × [1]

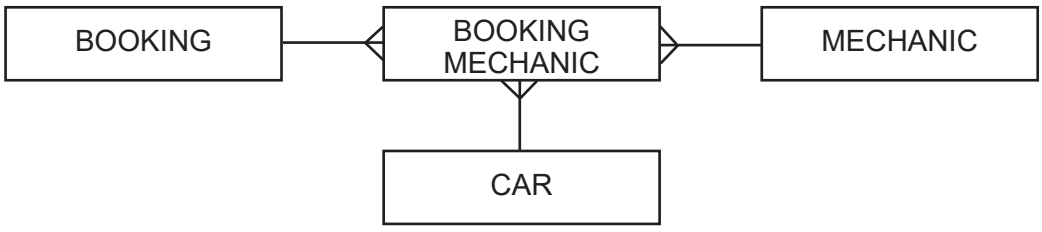
Primary key
Uniquely identifies
… a record in a table
2 × [1] [4]

(c) **(i)** **1NF**
BOOKING (**BookingID,** Date, Time, CustName)
BOOKING-MECHANIC**(BookingID, MechanicID,** MechName, CarReg, CarType)
2 × [1] [2]

(ii) **2NF**
BOOKING (**BookingID,** Date, Time, CustName)
BOOKING-MECHANIC**(BookingID, MechanicID,** CarReg, CarType)
MECHANIC(**MechanicID,** MechName)
3 × [1] [3]

(iii) **3NF**
BOOKING (**BookingID,** Date, Time, CustName)
BOOKING-MECHANIC**(BookingID, MechanicID,** CarReg)
MECHANIC(**MechanicID,** MechName)
CAR(**CarReg,** CarType)
4 × [1] [4]

(iv)



[1] for BOOKING-MECHANIC entity
3 × [1] for correct relationships
[1] for all three remaining entities [5]

(d) **(i)** INSERT INTO Employee (EmployeeID,EmployeeName,Position,Department) [1] + [1]
VALUES ('720','White', 'Tester','IT') [1] + [1] [4]

(ii) Delete from Employee [1] + [1]
where EmployeeID=655 [1] [3]

(iii) UPDATE Employee [1] + [1]
SET Position = 'Senior Programmer' [1] + [1]
WHERE EmployeeID = 652; [1] [5]

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AVAILABLE MARKS

4 **(a)** The Turing Test aims to determine if a machine's behaviour is indistinguishable from a human's response
The test involves a player, a human and a machine
The player does not know which is the human and which is the machine
The player asks questions of the human and the machine
… and tries to discover which is the human
4 × [1] [4]

(b) Models the structure of the human brain/neurons
… using a collection of nodes/units arranged in layers
The nodes/units are linked to each other by one-way/two-way connections
Some nodes/units are also connected to the outside world
Some nodes/units are connected to the outside world/provide the input/output for the network
Some nodes are hidden
Each connection has a weight/the sum of its inputs
… which represents its output/what it has 'learned' from the inputs
Backward propagation may be used
Algorithms are used to calculate the weights using data from relevant real-world examples
4 × [1] [4]

(c) Interrogates the knowledge base
… and draws inferences/conclusions
… based on the stored rules
It asks the user questions
… to determine a suitable response
4 × [1] [4]

(d) **(i)** **Medical consultants**
They will provide the knowledge
… during the knowledge acquisition stage
... by describing their experiences/knowledge of medical treatments
... including facts
… and heuristics
3 × [1]

An expert system shell
This is the software development environment for creating an expert system
It is an 'empty' expert system
… without the knowledge base for any particular application
It consists of a knowledge acquisition system/knowledge base/inference engine/user interface
It is populated/configured for the particular application
… with appropriate rules and facts
3 × [1]

The knowledge engineer
Collects the information/knowledge for the knowledge base
… from the experts/medical consultants
... using structured/unstructured interviews/problem solving/concept maps
Formulates this knowledge as a set of rules/facts
Validates/verifies the knowledge
3 × [1] [9]

AVAILABLE MARKS

(ii) Banded response

Description of the use of an expert system in medicine

The knowledge base contains facts and rules about a wide range of illnesses gathered by the knowledge engineer from medical consultants/ doctors/physicians

The user interface will query the patient's symptoms

The inference engine will apply the rules to the user's responses and suggest a range of diagnoses

Evaluation

The knowledge base will contain the knowledge of many medical experts, possibly from around the globe, which individual doctors cannot have

The expert system will justify its conclusions about its diagnoses which will support timely and effective decision making

The knowledge base can be regularly updated with information about new illnesses and treatments

Patients can use the expert system themselves, freeing up valuable resources

This use of an expert system may lead to deskilling among medical practitioners

An incorrectly designed expert system may produce incorrect results which may be difficult to detect and may have serious medical consequences

Level	Marking Criteria	Marks
Band 2	The candidate • Provides a detailed description of this use of an expert system in medicine • Provides an evaluation, with justification, of this use of an expert system referring to a benefit and a drawback • Uses the appropriate Digital Technology terminology accurately throughout the response Presentation, spelling, punctuation and grammar are of a high standard.	[5]–[6]
Band 1	The candidate • Provides a detailed description of this use of an expert system in medicine • Describes an advantage or disadvantage of this use of an expert system • Uses some relevant Digital Technology terminology Presentation, spelling, punctuation and grammar are sufficiently competent to make the response clear.	[3]–[4]
Band 0	The candidate • Provides a description of the use of an expert system in medicine which lacks detail **or** States an advantage or disadvantage of this use of an expert system • Makes limited use of Digital Technology terminology Presentation, spelling, punctuation and grammar are such that the intended meaning is not completely clear.	[1]–[2]

[6]

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AVAILABLE MARKS

5 (a) Cells

Hexagonal areas
… with their own mast
… and base station
They are strategically placed to ensure maximum coverage
3 × [1]

Handoff

When a network automatically
…. switches coverage responsibility
…. from one base station to another
… as a user making a call moves between cells
3 × [1] [6]

(b) The combination of a range of disciplines
Computer science/Mechanical engineering/Electrical engineering
To support the design, production and operation of robots
Incorporates sensors and feedback
4 × [1] [4]

(c) Robots can work in dangerous/inaccessible conditions
… and be operated by remote control
… and provide automatic feedback/sensors/cameras
… without putting humans at risk
2 × [1]

They can be programmed/reprogrammed
… to perform repetitive/complex tasks
… reliably/consistently/accurately/operate 24/7/without supervision
… improving productivity
2 × [1]

They can be miniaturised/built from robust materials/mass-produced
… to withstand extreme conditions
… to perform tasks impossible for humans
2 × [1] [6]

AVAILABLE MARKS

(d) Banded response

Description how cloud computing is used

The organisation's data will be managed by a third party

… to provide data storage, email and to perform backups

The data will be stored on a global network of servers/data farms on the Internet.

Evaluation: This use of cloud computing wrt data security

The third party is responsible for data security. The organisation is reliant on the third party for data security.

With cloud computing, most data access involves data transmission over the Internet which increases the risk of unauthorised access.

Level	Marking Criteria	Marks
Band 2	The candidate • Provides a detailed description of cloud computing which is correct • Provides a detailed evaluation of cloud computing, with justification, with respect to data security • Uses the appropriate Digital Technology terminology accurately throughout the response Presentation, spelling, punctuation and grammar are of a high standard.	[5]–[6]
Band 1	The candidate • Provides a detailed description of cloud computing which is correct **or** Discusses the impact on data security of using cloud computing • Uses some relevant Digital Technology terminology Presentation, spelling, punctuation and grammar are sufficiently competent to make the response clear.	[3]–[4]
Band 0	The candidate • Provides a description of cloud computing which lacks detail **or** Refers briefly to data security • Makes limited use of Digital Technology terminology Presentation, spelling, punctuation and grammar are such that the intended meaning is not completely clear.	[1]–[2]

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				AVAILABLE MARKS
6	(a)	(i)	Unauthorised access to computer material with intent to commit or facilitate commission of further offences Unauthorised modification of computer material 2 × [2] [4]	
		(ii)	The organisation's CMA/AU policy should ensure … that employees are aware of their obligations under the Act IT use by employees should be audited regularly … and suspect access activity investigated The organisation should implement a username and password system … to control/restrict employee access to computer systems 6 × [1] [6]	
	(b)	(i)	The Act was designed to protect the "intellectual property" rights of those individuals and organisations … that create and produce material based on original ideas 2 × [1] [2]	
		(ii)	The Act makes it illegal to copy, modify or distribute software … without the relevant permission It is automatically applied from the time the work is created Software piracy involves the illegal copying, modifying or downloading of software Organisations must ensure that all employees are aware of the terms of the Act … and the consequences of being in breach of it Employees must not be allowed to use unauthorised software Organisations must ensure they have appropriate multi-user licences The organisation must carry out audits on the copyright status of all software that it uses … and monitor who has access to that software 6 × [1] [6]	18
			Total	**150**